Working with Okta and SAML integration

Zoe Reid (She/Her)

Why

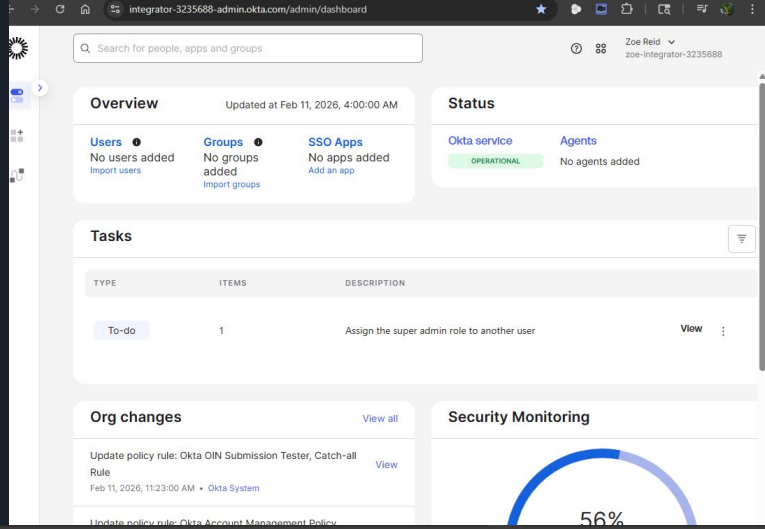
- Showcasing Blue Team things
- Delving deeper into IAM
- RDD
- Everything went wonky

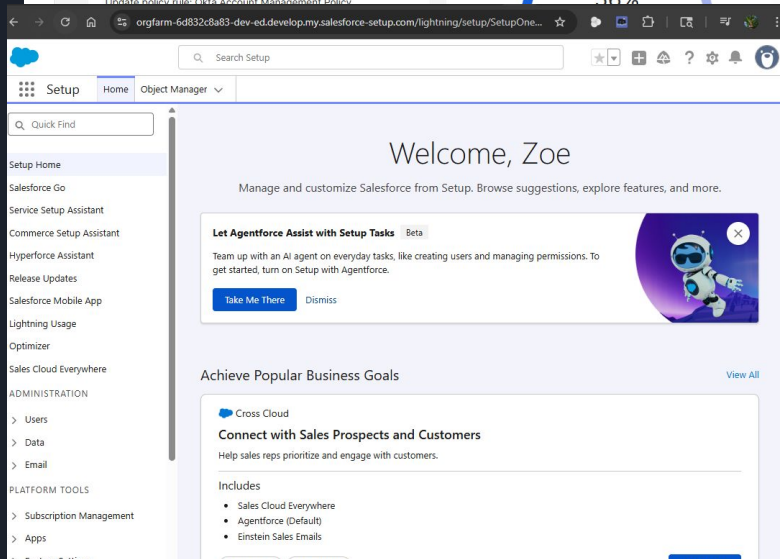
Background

- SAML (Security Assertion Markup Language)
- OIDC
- OAUTH 2.0

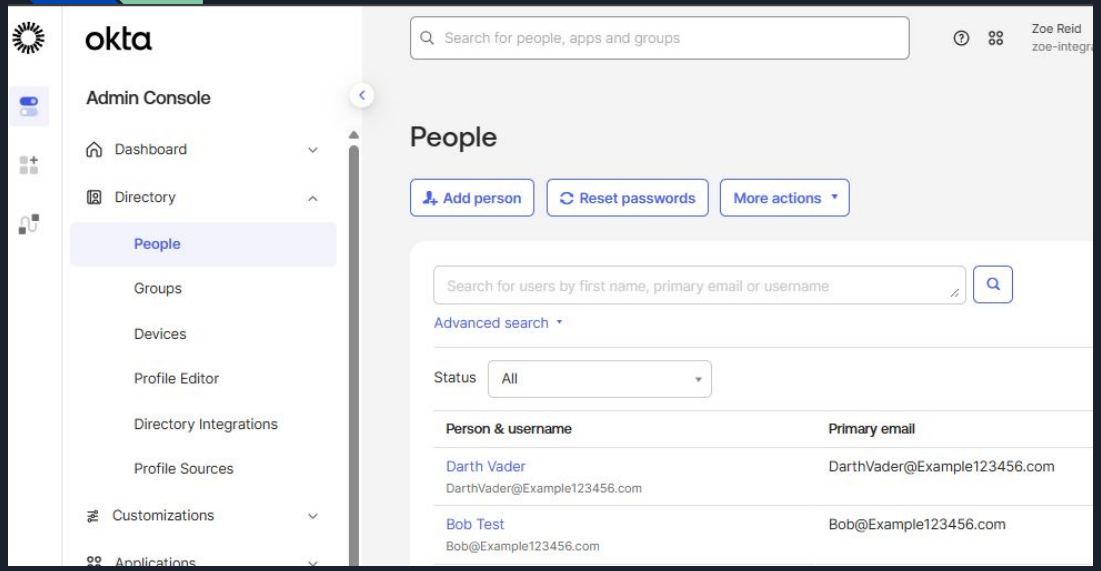
Setup

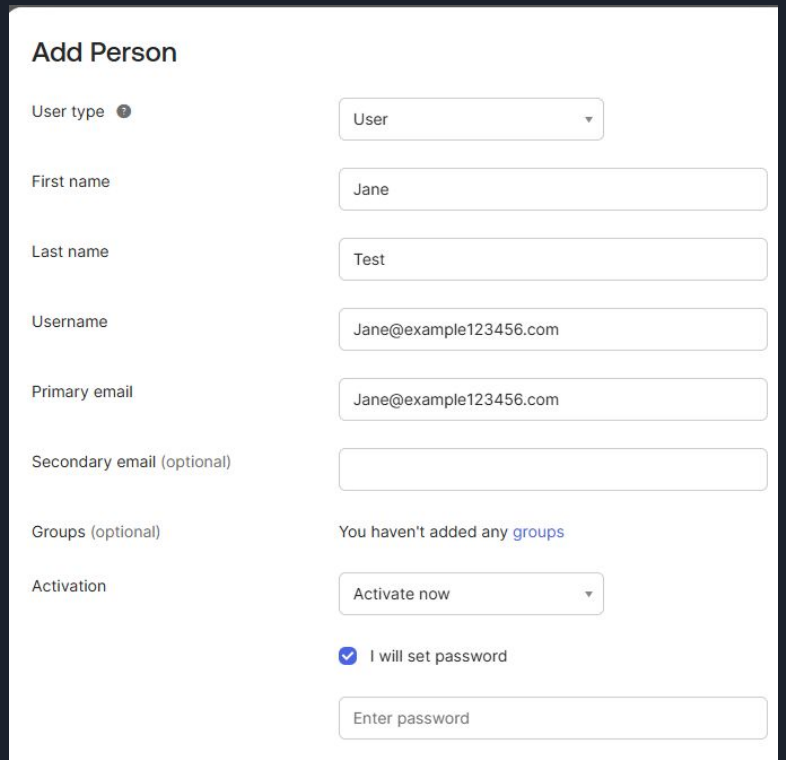
- Okta Developer Account
- Salesforce Developer Account
- Cost about $10 for org email





Adding Users in Okta





Setting Policies for Account Logins

integrator-3235688-admin.okta.com/admin/authn/authentication-policies#authentic...

Search for people, apps and groups

Zoe Reid
zoe-integrator-3235688

← Back to all Authentication Policies

Okta Account Management Policy

Documentation

This policy protects authenticator enrollment and unenrollment, changing a password from account settings, and editing personal information.

When password policy rules determine, this policy can also protect changing an expiring password, self-service password reset, and unlock account.

This policy does not protect admin-initiated password reset with an email link.

Confirm that users coming through authenticator enrollment policies and password policies can satisfy this policy. Don't require authentication with an authenticator that a user doesn't have.

Add rule

Priority	Rule	Status	Actions
1	Password Expiry Rule	ENABLED	Actions
	IF Expression: Yes THEN Access: Allowed with password Your org's authenticators that satisfy this requirement: Password Authentication methods: Allow any method that can be used to meet the requirement Re-authentication frequency is: Every time user signs in to resource		
2	Catch-all Rule	ENABLED	
	IF Any request THEN Access: Allowed with progressive authentication Users with only one factor enrolled will be required to authenticate with any one factor. Users with multiple factors		

Add Rule

If all of the conditions are true, the authentication settings below will apply. Otherwise, Okta will evaluate the next rule.

Rule name

IF

IF User's user type is — Any user type

AND User's group membership includes — Any group

AND User is — Any user

AND Device platform is — Any platform

Device platform is securely verified for registered devices. For unregistered devices, it's determined by the user-agent and can be modified. Learn more about Device platform security

AND User's IP is — Any IP

AND Risk is — Any

AND The following custom expression is true

This is an optional advanced setting. If the expression is formatted incorrectly or conflicts with conditions set above, the rule may not match any users.

Expression language reference

THEN

THEN Access is
- Denied
- Allowed after successful Authentication

Connecting Salesforce from Okta

Search for people, apps and groups

Add Salesforce.com

1 General Settings | 2 Sign-On Options

General settings· Required

Application label: Salesforce.com

This label displays under the app on your home pa

Instance Type: Production

Select the type of Salesforce instance that you wa connect to.

Custom Domain:

Enter your custom domain. If your domain is `acme.my.salesforce.com`, enter `acme`. This field is required if you are using your custom domain for y Entity Id in Salesforce or if you are using Salesforc Government Cloud. (How to Configure)

User Profile & Type: Standard Salesforce User

Select the type of user you wish to provision to or from Salesforce. "Standard" will create normal Sale users and is the default. If you have a Portal or Cor in Salesforce then you can instead choose to have provision these as "external users". Use multiple Salesforce.com application instances to handle bo "Standard" users as well "Portal" and "Community"

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integrator-3235688-admin.okta.com/admin/app/salesforce/instance/_new_/

Search for people, apps and groups

Add Salesforce.com

1 General Settings | 2 Sign-On Options

Sign-On Options· Required

Sign on methods

The sign-on method determines how a user signs into and manages their credentials for a application. Some sign-on methods require additional configuration in the 3rd party applic

Application username is determined by the user profile mapping. Configure profile mappin

- Secure Web Authentication
- Bookmark-only
- SAML 2.0

Default Relay State:

All IDP-initiated requests will include this RelayState.

Attributes (Optional) Learn More

Maximum App Session Lifetime: Send value in response

Uses SessionNotOnOrAfter attribute

Max limit 90 Days

Preview SAML

4 Check the **SAML Enabled** box to enable the use of SAML Single-Sign On, then click **Save**:

Single Sign-On Settings

Save | Cancel

Federated Single Sign-On Using SAML

SAML Enabled

Save | Cancel

5 Click **New**:

Single Sign-On Settings

Configure single sign-on in order to authenticate users in salesforce.com from external environments. Your organization has the following options available for single sign-on:

- Federated authentication, a single sign-on method that uses SAML assertions sent to a Salesforce endpoint.

Edit | SAML Assertion Validator

Federated Single Sign-On Using SAML

SAML Enabled ✓

SAML Single Sign-On Settings: New | New from Metadata File | New from Metadata URL

6 Enter the following:

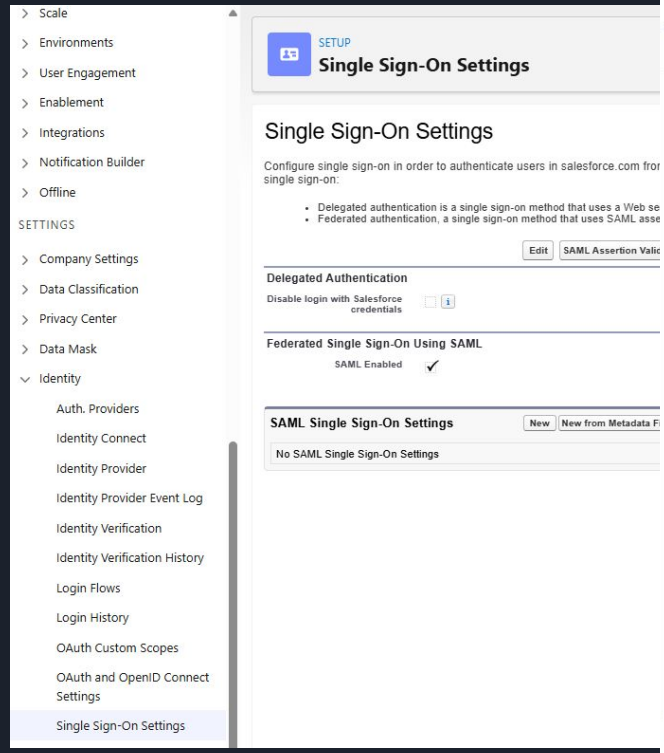
Unless otherwise noted, leave the default values as-is.

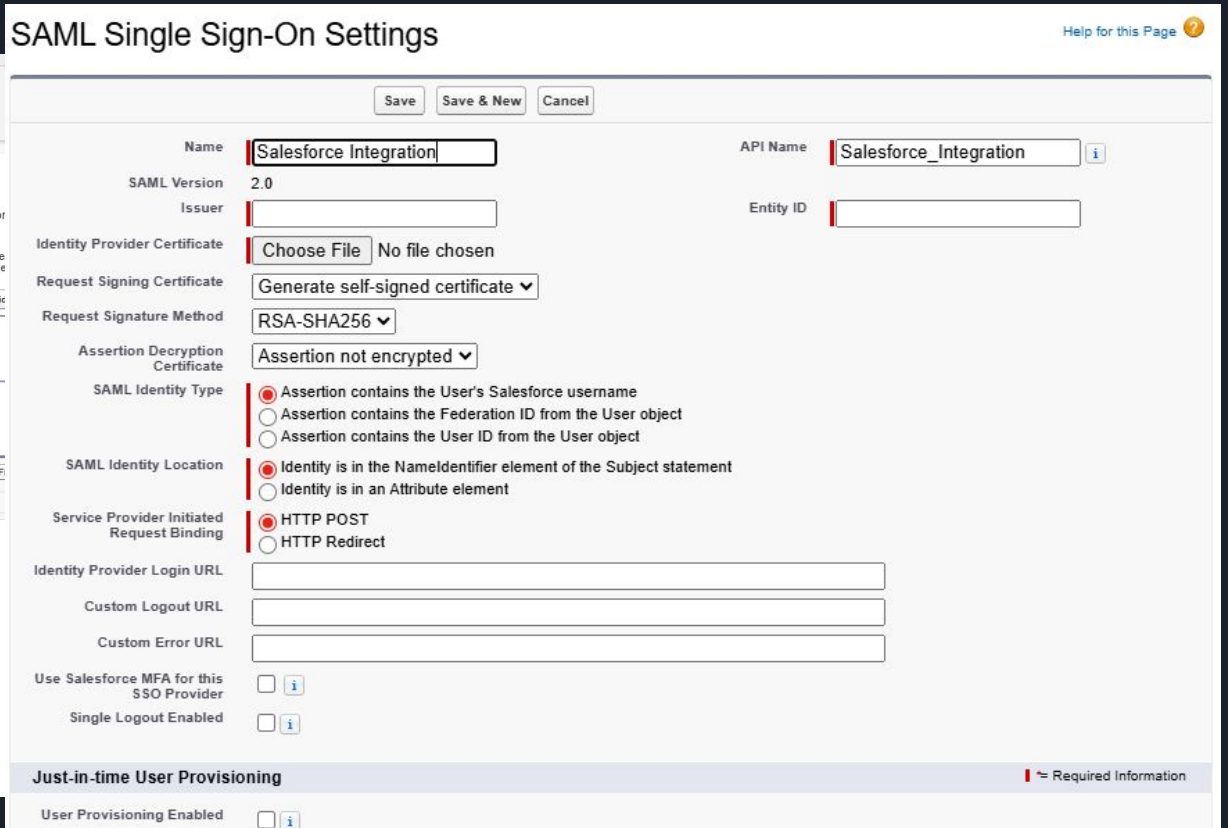
- **Name**: Enter a name of your choice.
- **SAML Version**: Make sure this is set to **2.0**. This should be enabled by default.
- **Issuer**: Copy and paste the following:

```
exk101wa2d7APzPSG698
```

- **Identity Provider Certificate**: Download, then upload the following certificate into this field:

Connecting Salesforce from the Salesforce Side





Salesforce Completed SSO Config

SETUP

Single Sign-On Settings

SAML Single Sign-On Settings

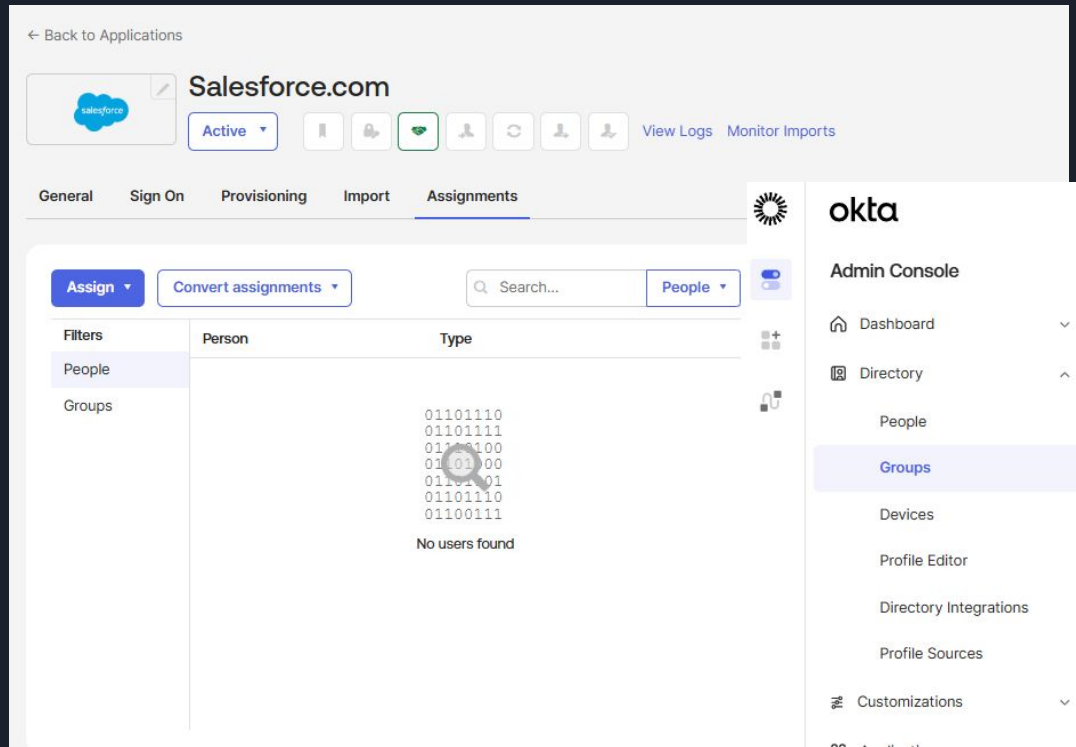
Help for this Page

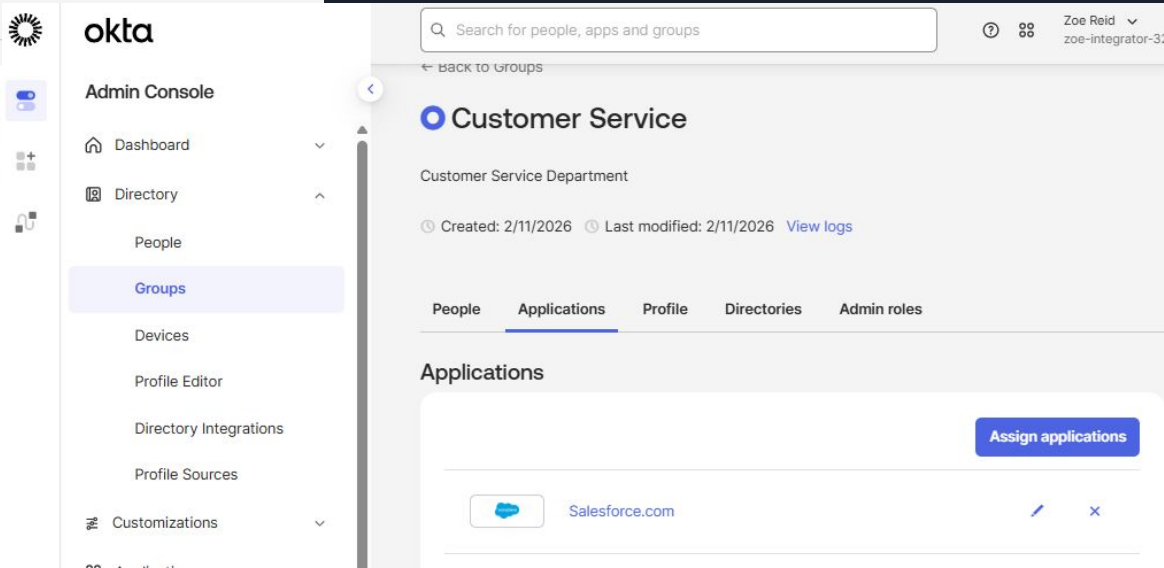
Back to Single Sign-On Settings

Edit | Delete | Clone | Download Metadata | SAML Assertion Validator

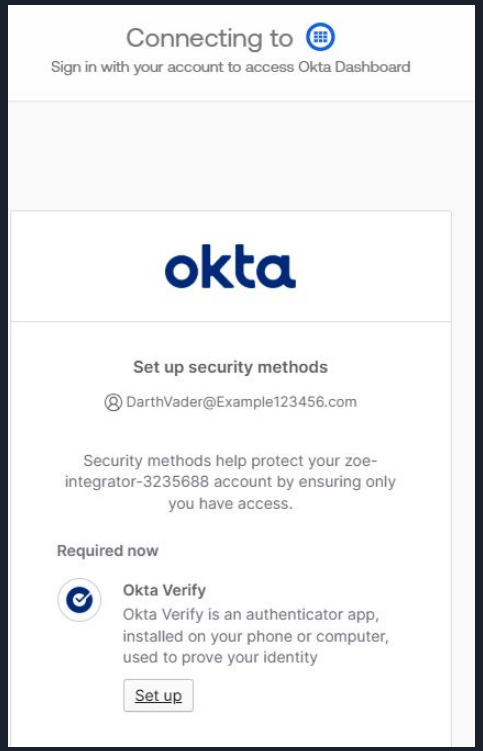
Name	Okta	API Name	Okta
SAML Version	2.0		
Issuer	exk101ws2d7APzPSG698	Entity ID	https://saml.salesforce.com
Identity Provider Certificate	EMAILADDRESS=info@okta.com, CN=integrator-3235688, OU=SSOProvider, O=Okta, L=San Francisco, ST=California, C=US Expiration: 11 Feb 2036 23:28:51 GMT		
Request Signing Certificate	SelfSignedCert_11Feb2026_234216		
Request Signature Method	RSA-SHA256		
Assertion Decryption Certificate	Assertion not encrypted		
SAML Identity Type	Username		
SAML Identity Location	Subject		
Service Provider Initiated Request Binding	HTTP POST		
Identity Provider Login URL	https://integrator-3235688.okta.com/app/salesforce/exk101ws2d7APzPSG698/sso/saml		
Custom Logout URL	https://integrator-3235688.okta.com		
Custom Error URL			
Use Salesforce MFA for this SSO Provider	☐		
Single Logout Enabled	☐		

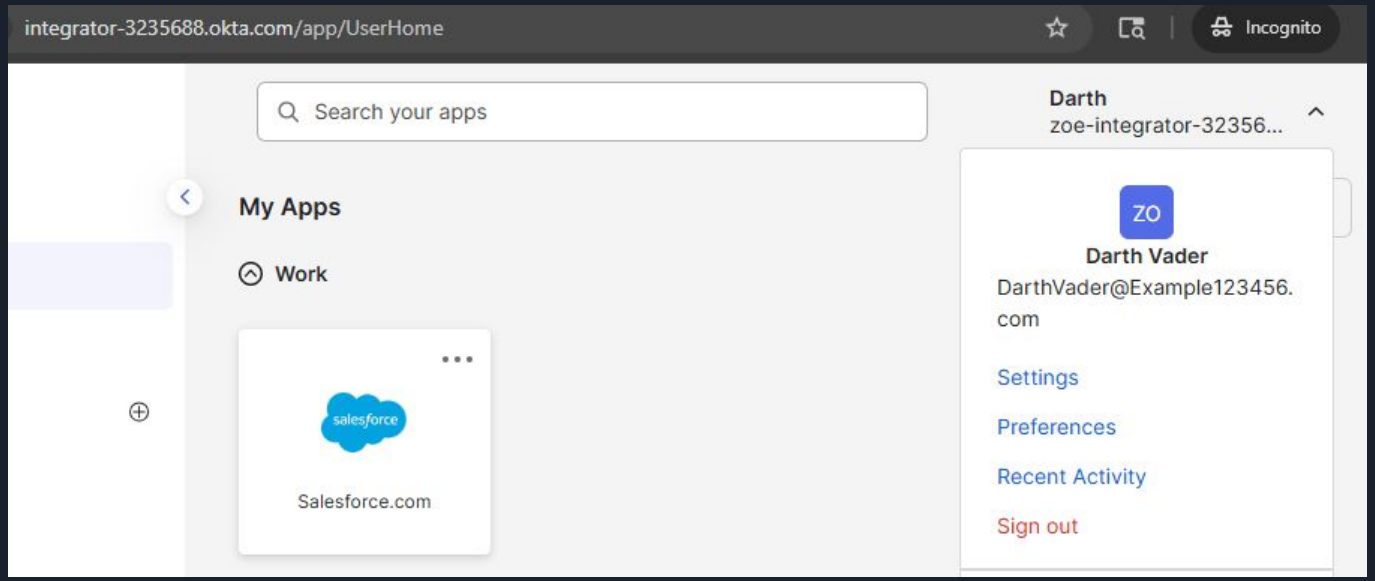
Completed App Integration in Okta





Verifying Correct Setup





SSO Error

orgfarm-6d832c8a83-dev-ed.develop.my.salesforce.com/_nc_external/identity/saml/SamlError

salesforce

Single Sign-On Error

We can't log you in because of an issue with single sign-on. Contact your Salesforce admin for help.

SETUP
Users

All Users

Help for this Page

On this page you can create, view, and manage users.

To get more licenses, use the Your Account app. Let's Go

View: All Users | Edit | Create New View

A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | Other | All

New User | Reset Password(s) | Add Multiple Users

Action	Full Name ↑	Alias	Username	Role	Active	Profile
Edit	Chatter Expert	Chatter	chatty.00dgl00000kerchuav.rs4smu9gov26@chatter.salesforce.com		✓	Chatter Free User
Edit	EPIC, OrgFarm	OEPIC	epic.0dca6e88ee03@orgfarm.salesforce.com		✓	System Administrator
Edit	Reid, Zoe	zjr	zjreid.0a026c71b7e4@agentforce.com		✓	System Administrator
Edit	User, Integration	integ	integration@00dgl00000kerchuav.com		✓	Analytics Cloud Integration User
Edit	User, Security	sec	insightssecurity@00dgl00000kerchuav.com		✓	Analytics Cloud Security User

New User | Reset Password(s) | Add Multiple Users

A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | Other | All

SETUP
Users

New User

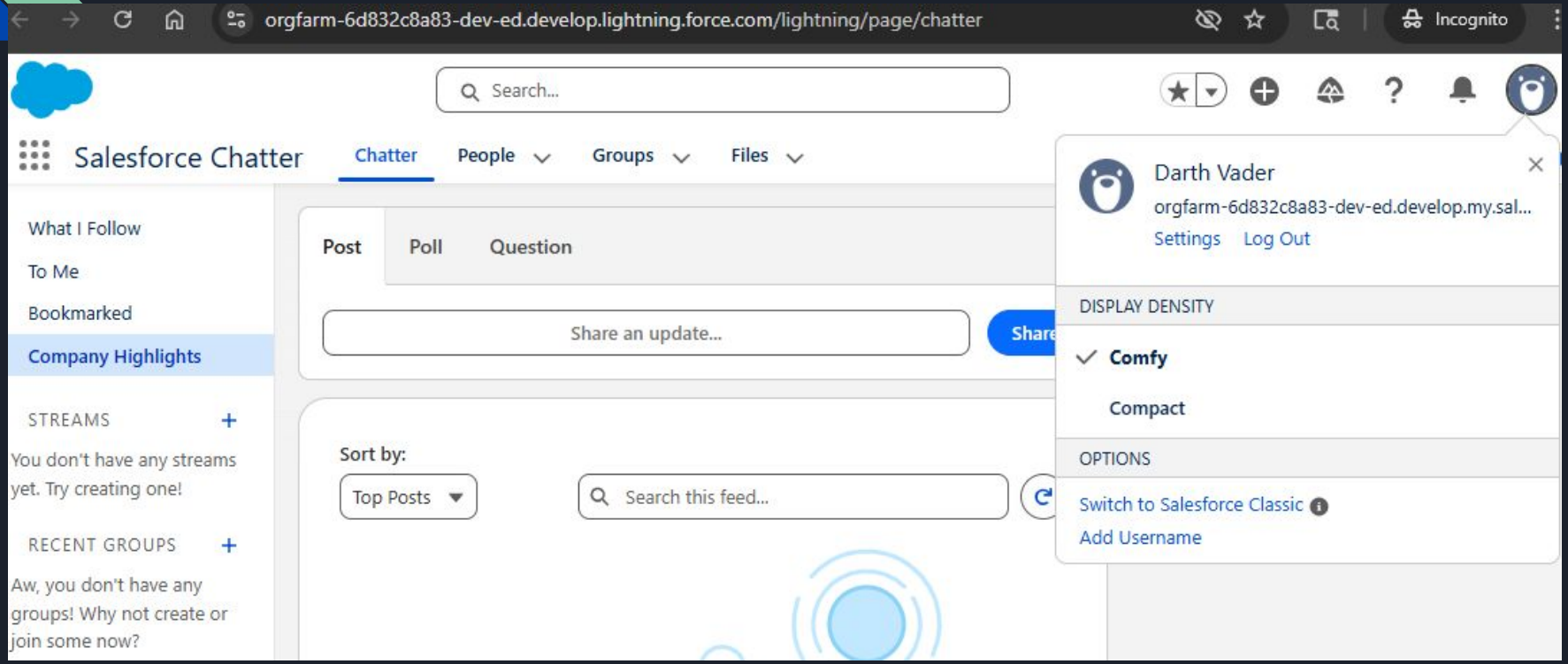
Help for this Page

User Edit Save | Save & New | Cancel

General Information Required Information

First Name	Darth	Role	<None Specified>
Last Name	Vader	User License	Chatter Free
Alias	dvade	Profile	Chatter Free User
Email	DarthVader@Example12345	Active	✓
Username	DarthVader@Example12345	Marketing User	
Nickname	User1770855560929568693	Offline User	
Title		Knowledge User	
Company		Flow User	
Department		Service Cloud User	
Division		Site.com Contributor User	

Success


orgfarm-6d832c8a83-dev-ed.develop.lightning.force.com/lightning/page/chatter
Incognito
Search...
Salesforce Chatter
Chatter
People
Groups
Files
What I Follow
To Me
Bookmarked
Company Highlights
STREAMS
You don't have any streams yet. Try creating one!
RECENT GROUPS
Aw, you don't have any groups! Why not create or join some now?
Post
Poll
Question
Share an update...
Share
Sort by:
Top Posts
Search this feed...
Darth Vader
orgfarm-6d832c8a83-dev-ed.develop.my.sal...
Settings
Log Out
DISPLAY DENSITY
Comfy
Compact
OPTIONS
Switch to Salesforce Classic
Add Username

Verifying with Logs

Search for people, apps and groups

Zoe Reid
zoe-integrator-3235

System Log

← Back to Reports

From 02/05/2026 00:00:00 To 02/12/2026 23:59:59 (GMT-8) Pacific Time - America/Los_Angeles

Search

actor.id eq "00u101w73ccGHwJ4f698" or target.id eq "00u101w73ccGHwJ4f698" Save

Advanced Filters / Reset Filters

Count of events over time

Show event trends by category

Events: 27 Download CSV

	Time	Actor	Event Info	Targets
	Feb 11 16:21:30	Darth Vader (User) 67.172.174.141	User single sign on to app SUCCESS	Salesforce.com (AppInstance) Darth Vader (AppUser)
	Feb 11 16:21:30	Darth Vader (User) 67.172.174.141	Evaluation of sign-on policy ALLOW	Salesforce.com (AppInstance) Catch-all Rule (Rule) 1 more targets
	Feb 11 16:16:40	Darth Vader (User) 67.172.174.141	User single sign on to app SUCCESS	Salesforce.com (AppInstance) Darth Vader (AppUser)
	Feb 11 16:16:39	Darth Vader (User) 67.172.174.141	Evaluation of sign-on policy ALLOW	Salesforce.com (AppInstance) Catch-all Rule (Rule)

SETUP
Login History

Login History

Help for this Page

Download the last six months of login history. Or, filter and show up to 20,000 records of login history for the last six months.

Location shows the approximate location of the IP address from where the user logged in. To show more geographic information, such as approximate city and postal code, create a custom view to include those fields. Due to the nature of geolocation technology, the accuracy of geolocation fields (for example, country, city, postal code) may vary.

Download Options

File Type
- CSV File
- GZIP File

File Contents: All Logins

Download Now

View: All Create New View

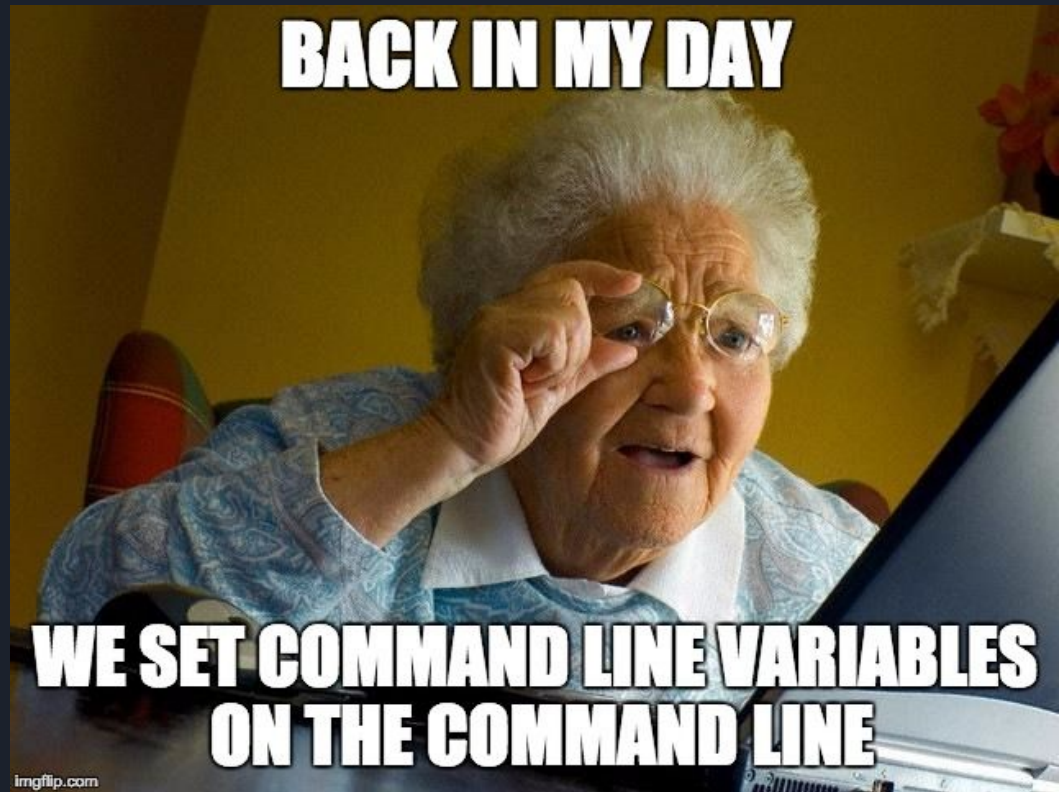
Username	Login Time ↓	Source IP	Location	Login Type	Status	Browser	Platform	Application	Client Ver
darthvader@example123456.com	2/11/2026, 4:21:30 PM PST	67.172.174.141	United States	SAML Idp Initiated SSO	Success	Chrome 144	Windows 10	Browser	N/A

Lessons Learned

- Make sure you have solid documentation
- It’s easy to get lost in config details
- When troubleshooting, analyze the problem from all sides

Future Work/Integrations

- Connecting AD
- Keycloak
- Automation for Joiner/Mover/Leaver scenarios



Questions